

Garv Arora

Computer Science Engineer | Intelligent Robotic Systems & Software

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Summary

Computer Science engineer specializing in ROS2 navigation, real-time sensor fusion, and ARM bare-metal firmware — shipping competition-grade software for planetary rover and drone platforms. Placed 13th (IRC 2025) and 17th (IRC 2026) globally among 100+ international teams. Experienced across CUDA-accelerated perception pipelines, multi-agent LLM systems, and microcontrollers. Looking for roles in software engineering, robotics, embedded systems, or ADAS.

Education

Vellore Institute of Technology (VIT)

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36 / 10.0

Gems International School (CBSE)

Class XII: 76.4% | Class X: 89.3%

Vellore, India

August 2023 – July 2027

Gurugram, India

November 2016 – April 2023

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog

Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID Control, Sensor Fusion, mmWave Radar

Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers

Embedded & Infrastructure: ARM Cortex-M, ESP32, I2C/SPI/UART, FreeRTOS, Linux Kernel/D-Bus, Docker, Git, GitLab CI/CD, AWS, PyQt, PostgreSQL

Work Experience

SEDS Projects VIT | SEDS India

April 2024 – Present

Research & Development Lead (Promoted from Developer)

January 2026 – Present

- Directing 2 space robotics competition teams (50+ engineers) — Team Ardra (ISDC) and Team Vyadh (IRC) — under SEDS India, conducting architecture reviews and debugging, resolving 10+ critical pre-competition issues.

Autonomous Systems Developer

April 2024 – January 2026

- Architected ROS2 Nav2 & RTAB-Map 3D Visual SLAM pipeline on Intel RealSense D455, generating real-time elevation maps and slashing manual teleoperation time by 70% in competition.
- Slashed CV pipeline latency by ~60% and raised throughput 4x via CUDA, YOLOv8, ArUco/AprilTag tracking, and Linux V4L2 zero-copy buffer tuning across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware to ESP32 microcontrollers for distributed data acquisition; fused wheel encoders and IMU data with an EKF, cutting odometry drift by ~40% (sub-10 cm positioning accuracy).
- Shipped PyQt5 autonomous mission dashboard with real-time telemetry visualization, parameter tuning, and FSM-based mission sequencing for autonomous competition runs.

Samsung PRISM

Remote

Data Intelligence Agent Developer (Internship)

August 2025 – Present

- Engineered multi-agent LangGraph data engine (FastAPI, Redis, PostgreSQL, Qdrant) automating discovery, screening, and 20-stage analysis of dataset corpora; integrated spaCy NLP, Presidio PII anonymization, and MemorySaver HITL pause.

LG Soft India (LGSi) – HS/ES India Lab

Bengaluru, India

Living Solution Control Development Intern

June 2026

- Constructed C/C++ embedded control logic and state machine handlers for living solution control modules, verifying execution timing on target hardware; work certified as “Technically Competent” by LGSi HR/Engineering management.

Wissen Baum Engineering Solutions LLP

Pune, India

Software Automation Intern

May 2025 – July 2025

- Authored Python BDD test automation framework (Gherkin/Behave, pytest, Playwright), parallelizing GitLab CI jobs to cut pipeline execution time from ~10m to ~6m (40% speedup) and reducing manual validation effort by 80%+.

Projects

Autonomous Radar Survivor Detection (Patent App IN202641072249 A1) | ROS2, Hi-Link LD2410C, ESP32

2026

- Interfaced an LD2410C 24GHz FMCW mmWave radar with ROS2 to detect survivor breathing through 30 cm of rubble; fused radar point-cloud with IMU odometry, formulating confidence grid decay $C = \max(0, C - \beta \Delta t)$ for survivor mapping.

HayaiOS – Bare-Metal RTOS Kernel | C, ARM Assembly, ARM Cortex-M4

2026

- Implemented preemptive scheduler (1 ms tick), context switching in <50 assembly instructions, mutex/semaphore primitives, and a full HAL isolating all register-level I/O on ARM Cortex-M4.

Captivity CLI | Rust, Python, Linux D-Bus, systemd, Socket IPC | pypi.org/project/captivity-cli

2026

- Built autonomous captive portal WiFi login daemon featuring <50ms HTTP probing, systemd background integration, D-Bus NetworkManager monitoring, keyring credential encryption, and dual Python-Rust TCP socket IPC.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and IMU via complementary filter, reducing tracking failures by ~50% across a 6-DOF fast-motion test trajectory.

Certifications & Achievements

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (~2.3% acceptance rate)

International Rover Challenge (IRC): Ranked **13th Globally (2025)** and **17th Globally (2026)** with Team Vyadh (SEDS VIT)

Certifications: AWS Certified AI Practitioner | AWS Certified Cloud Practitioner | Oracle Cloud Data Science & GenAI Professional